Lecture Notes on Data Science: k-Means Clustering

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This is the first in a series of lecture notes on k-means clustering, its variants, and applications. We discuss the basic ideas behind k-means clustering and study the classical algorithm.

Introduction

Cluster analysis is often applied to data that can be embedded in Euclidean vector spaces. Let us therefore assume a set of data

$$X = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\} \subset \mathbb{R}^m \tag{1}$$

whose n elements $\boldsymbol{x}_j$ are m dimensional real-valued vectors that are to be clustered (see Fig. 1 for an example).

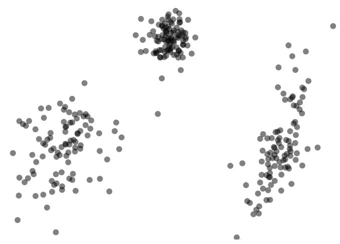


Figure 1: Didactic example of $n = 150$ data points $\boldsymbol{x}_j \in \mathbb{R}^2$ sampled from three bivariate Gaussian distributions.

When we attempt to cluster data, our intuitive goal is to detect latent structures or groups of similar data points within the given sample. Turning this intuition into a choice for a specific clustering algorithm requires us to think about the following questions:

Q1: what defines similarity?

Q2: which properties should a cluster have?

Unfortunately, there are numerous definitions of what constitutes a cluster and at least as many algorithms. Since different clustering algorithms are designed with respect to different answers to Q1 and Q2, we should never use them naïvely because not every algorithm applies to every problem. In this note, we therefore study the basic ideas behind the popular k-means algorithm which will later help us understand where and how to use it.

k-Means Clustering

Traditional k-means clustering is based on rather simple and straightforward answers to the questions raised above.

Q2 is answered in that the k-means algorithm represents a cluster in terms of a subset $C_i \subset X$ and attempts to partition the data into k clusters that meet the following criteria:

$$\text{clusters are pairwise disjoint, i.e. } C_i \cap C_j = \emptyset \text{ for } i \neq j \tag{2}$$

$$\text{clusters cover the data such that } C_1 \cup C_2 \cup \ldots \cup C_k = X \tag{3}$$

Regarding Q1, the idea employed in k-means clustering is to represent each cluster by means of a *centroid* $\boldsymbol{\mu}_i \in \mathbb{R}^m$ and to measure similarities in terms of Euclidean distances between centroids and data points. Hence, two data points are considered similar if their distances to a common centroid are small and a point $\boldsymbol{x}_j$ will be assigned to cluster C_i if its distance to $\boldsymbol{\mu}_i$ is less than that to any other centroid.

cluster centroids

THIS NOTION OF SIMILARITY reduces the clustering problem to the problem of finding appropriate centroids which can be formalized as minimizing the following *objective function*

$$E(k) = \sum_{i=1}^{k} \sum_{\boldsymbol{x}_j \in C_i} \left\| \boldsymbol{x}_j - \boldsymbol{\mu}_i \right\|^2 \tag{4}$$

k-means objective function

where minimization happens w.r.t. the $\boldsymbol{\mu}_i$. Thus, minimizing (4) is to determine suitable centroids $\boldsymbol{\mu}_i$ such that, if the data are partitioned into corresponding clusters C_i, distances between data points and their closest cluster centroid become as small as possible.

ALTHOUGH THE OBJECTIVE IN (4) LOOKS RATHER INNOCENT and formalizes an intuitive idea, finding an optimal solution, i.e. finding global minimizers $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_k$, is actually NP-hard[1]. Even for 2D data and settings where $k = 2$, it cannot be guaranteed that the overall best possible solution will be found in reasonable time. This is important to know because it tells us that the k-means algorithm is merely a heuristic for dealing with a surprisingly hard problem.

[1] D. Aloise, A. Deshapande, P. Hansen, and P. Popat. NP-Hardness of Euclidean Sum-of-Squares Clustering. *Machine Learning*, 75(2), 2009

The Classical k-Means Algorithm

So, how does the k-means algorithm proceed to minimize (4)? In a nutshell, it realizes a greedy iterative update scheme. When started, it randomly initializes the centroids $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_k$. Given these initial guesses, the data are clustered accordingly. That is, the algorithm determines k clusters where

$$C_i = \left\{ \boldsymbol{x}_j \in X \;\middle|\; \left\| \boldsymbol{x}_j - \boldsymbol{\mu}_i \right\|^2 \leq \left\| \boldsymbol{x}_j - \boldsymbol{\mu}_l \right\|^2 \; \forall \, l \neq i \right\} \tag{5}$$

Once clusters have been determined, the algorithm updates its current estimates of the cluster centroids by computing

$$\boldsymbol{\mu}_i = \frac{1}{n_i} \sum_{\boldsymbol{x}_j \in C_i} \boldsymbol{x}_j \tag{6}$$

where $n_i = |C_i|$ denotes the number of elements in cluster C_i. As these updates may lead to a new clustering, steps (5) and (6) are repeated until the assignment of data points to clusters stabilizes or a predefined number of iterations is exceeded (see Fig. 2).

$t \leftarrow 0$
done $\leftarrow$ False
randomly initialize $\boldsymbol{\mu}_1^t, \boldsymbol{\mu}_2^t, \ldots, \boldsymbol{\mu}_k^t$
while not *done* **do**
  // for each mean, update its cluster
  for $i = 1$ to k **do**
    $C_i^t = \left\{ \boldsymbol{x} \in X \,\middle|\, \left\| \boldsymbol{x} - \boldsymbol{\mu}_i^t \right\| \leq \left\| \boldsymbol{x} - \boldsymbol{\mu}_j^t \right\| \right\}$

  // for each cluster, update its mean
  for $i = 1$ to k **do**
    $\boldsymbol{\mu}_i^{t+1} = \frac{1}{|C_i^t|} \sum_{\boldsymbol{x} \in C_i^t} \boldsymbol{x}$

  // tests for convergence
  if $C_i^t \cap C_i^{t-1} = C_i^t \; \forall \, i$ **then**
    done $\leftarrow$ True

  if $\left\| \boldsymbol{\mu}_i^{t+1} - \boldsymbol{\mu}_i^t \right\| \leq \epsilon \; \forall \, i$ **then**
    done $\leftarrow$ True

  $t \leftarrow t + 1$
  if $t > t_{\max}$ **then**
    done $\leftarrow$ True

end while

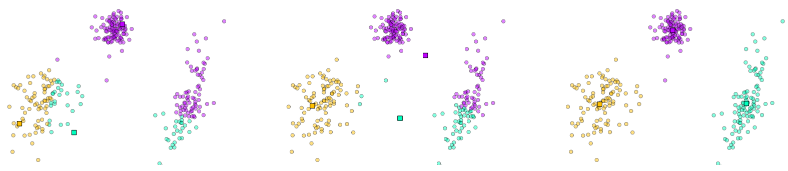
Figure 2: Pseudo code for k-means.

IT IS EASY TO SHOW that using (6) to update the centroids to the sample means of the clusters computed in (5) will never increase the value of the objective function in (4)[2]. Each iteration therefore improves on the previous result and the k-means algorithm usually converges quickly (see Fig. 3 for an example).

[2] We will prove this in a later note.

However, since the algorithm starts from a random initialization, it cannot be guaranteed to find the global minimum of its objective function. As it typically converges to a local minimum, it is pivotal to run k-means several times so as to empirically determine good clusters[3].

[3] Here is a video to illustrate this point: www.youtube.com/watch?v=9nKfViAfajY



(a) initialization (b) first update (c) final result

Figure 3: $k = 3$ means clustering of the data ($\circ$) in Fig. 1; (a) initial centroids ($\square$) and clustering, (b) situation after the first update, and (c) result upon convergence.

In this example, it took only few iterations to converge to a good solution. However, there is no guarantee for k-means to behave this way. One should therefore run it several times and base any further analysis on the result with the lowest value of the objective in (4).

Yet, even then, k-means may produce questionable results when seeded with inappropriate centroids. Intelligent initializations are the topic of ongoing research and arbitrary initialization are considered harmful; initial centroids should at least be selected among the available data points[4,5].

[4] P.S. Bradley and U.M. Fayyad. Refining Initial Points for K-Means Clustering. In *Proc. ICML*, 1998

[5] D. Arthur and S. Vassilvitskii. k-means++: The Advantages of Careful Seeding. In *Proc. SODA*, 2007

Notes and References

The origins of the idea of k-means clustering can be traced back to work by Steinhaus[6] in the 1950s.

The classical algorithm, i.e. the iterative procedure discussed in this note, was first discussed by Lloyd in a Bell Labs technical report also in the 1950s but openly published only much later[7].

Finally, the now so commonly used term "k-means" was coined by MacQueen[8] in the 1960s.

[6] H. Steinhaus. Sur la Division des Corps Matériels en Parties. *Bulletin Polish Acad. of Sciences*, 4(12):801–804, 1957

[7] S.P. Lloyd. Least Squares Quantization in PCM. *IEEE Trans. on Information Theory*, 28(2), 1982

[8] J.B. MacQueen. Some Methods for Classification and Analysis of Multivariate Observations. In *Proc. Berkeley Symp. on Mathematical Statistics and Probability*, 1967

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